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How We Make a Move on the Organic Ecological Chessboard

Summary

As conventional agriculture transitions to organic practices, balancing agricultural management measures with ecosystem stability has become increasingly urgent. This paper develops an integrated modeling framework focusing on these interactions. We establish several models and use various **visualization techniques** to make the results more intuitive.

In Model 1, to simulate seasonal variations among crops, weeds, pests, and predators, we integrate the **Lotka–Volterra equations** with the **Logistic Growth Model**. We adjust herbicide and pesticide usage using the method of controlling variables to assess their impact on crop yield and ecosystem stability. The findings indicate that when application intensity is too low, crop yield decreases significantly, though the ecosystem remains relatively stable. When application intensity is too high, pests are effectively suppressed, but the destructive effects of chemical agents on the farmland ecosystem become more pronounced, and the increase in crop yield is not particularly substantial .(*result: Figure 4*)

In Model 2, we incorporate **Cotton bollworm** and **peregrine falcon** to simulate the impact of returning native species on the ecosystem, and employ a **genetic algorithm**, based on pest control, ecosystem stability, and crop yield, to optimize chemical usage. Findings show that after native species return, the farmland ecosystem becomes **428%** more stable, crop yield increases by **6.6%**, and maintenance costs decrease by **5.2%**. Also, pests are effectively controlled, reliance on chemical use is minimized, and the ecosystem's resilience is enhanced.

In Model 3, we remove herbicides and introduce **bees** (pollinators) to study their effects on the ecosystem. We integrate the **pollination effects** of bees and bats into the crop growth model and use a **genetic algorithm** to optimize **pesticide application** before and after bees are introduced. The results indicate that removing herbicides lowers crop yield but raises ecosystem stability. After introducing pollinators, both the optimal pesticide dose and total costs decrease, while ecosystem stability rises by **65.91%**.

In Model 4, **to simulate various agricultural scenarios, we combine multiple organic farming strategies, including applying organic fertilizers, physical weeding, physical deworming, and biocontrol**. A **Pareto optimization** framework is employed to evaluate the advantages and disadvantages of each strategy in terms of **cost, farmland status, and ecosystem stability**. We then visualize different strategies using a **Cellular Automaton**, providing recommendations based on real-world conditions. (*result: Table 5*). Due to the model results, we compose an advisory letter for farmers transitioning to green agriculture, offering scientifically informed agricultural practices.

A sensitivity analysis examines the effect of fertilization intensity on ecosystem stability and farmland condition indicators, revealing very small sensitivity coefficients. We also perform a **regression analysis** to assess the effects and interactions of various measures on ecosystem stability. Finally, we discuss the strengths and limitations of the model.

Keywords:Logistic Growth Model, Lotka-Volterra Equation, Genetic Algorithm, Organic Agriculture, Pareto optimization, Cellular Automaton, organic farming strategies

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1. Introduction

1.1 Problem Background

Deforestation in the Brazilian Amazon is driven by the expansion of agricultural and grazing lands, leading to soil degradation and attracting widespread international attention. In this scenario, fertile forests are converted into farmland, disrupting ecological balance. Farmers frequently apply chemical herbicides and pesticides to boost yields, while effective against pests, harm non-target species beneficial to the ecosystem, such as bats and birds. Addressing these issues requires analyzing how the return of native species and the use of chemical agents impact the new agricultural ecosystem over time. Such analysis can inform strategies for farmers to enhance the sustainability and productivity of organic farming practices, achieving a balance between agricultural development and ecological conservation.

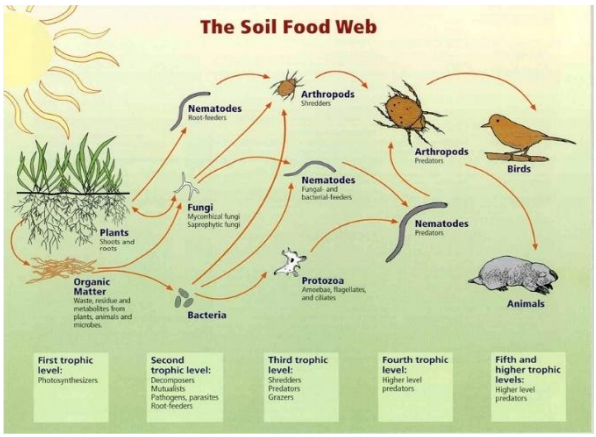


Figure 1: A Soil food web



Figure 2: Farmland with grazing

Figure 1 illustrates a typical soil food web, encompassing organisms from various trophic levels. It highlights the energy flow and interactions among species, showcasing the biodiversity and complexity characteristic of a healthy ecosystem.

Figure 2 depicts a pastureland grazed by sheep, with the stark contrast between the forest in the background and the farmland underscoring ecological transformation. This emphasizes the shift from natural equilibrium to human intervention. Furthermore, the grazing activity suggests potential impacts of agricultural practices on the original ecosystem.

1.2 Restatement of the Problem

Considering the background information and constraints defined in the problem statement, we need to address the following issues:

- **For Task 1**, we need to construct a food web model of the farmland ecosystem. This considers producers and consumers while accounting for the seasonal variations in the agricultural cycle. Additionally, we must analyze the impact of herbicides and pesticides on target species such as insects and weeds, as well as on non-target species like crops, bats, birds, to find the overall stability of the ecosystem.
- **For Task 2**, we need to optimize a new model based on Task 1 by introducing two new native species and quantitatively analyze their reintroduction's impact on three parameters:

pest control in the agricultural ecosystem, ecosystem stability, and crop yield. We also need to examine the interactions between the reintroduced native species and producers, consumers, and other species.

• **For Task 3**, we need to build on the model from Task 2 by removing herbicides and introducing pollinating bats and another species as crop pollinators. Then analyze how the changes affect pest control, ecosystem stability, and crop yield.

• **For Task 4**,we need to explore scenario analysis for transitioning to organic farming by introducing components like organic fertilizers, physical weeding, deworming, and biocontrol. An optimization model will evaluate the impact of these measures on the ecosystem, with multi-objective optimization assessing ecological and economic benefits.

1.3 Our work

Our work builds upon existing literature, including foundational ecological models such as the Lotka-Volterra model and modern techniques like genetic algorithms and Pareto optimization. While previous studies have primarily focused on isolated aspects of farmland ecosystems, such as species competition or chemical impacts, we extend this by constructing a comprehensive framework. This framework not only integrates the dynamic interactions between natural processes (e.g., species reemergence, food webs) and human interventions (e.g., herbicide reduction, organic farming practices) but also emphasizes multi-objective optimization to balance ecological stability and agricultural productivity.

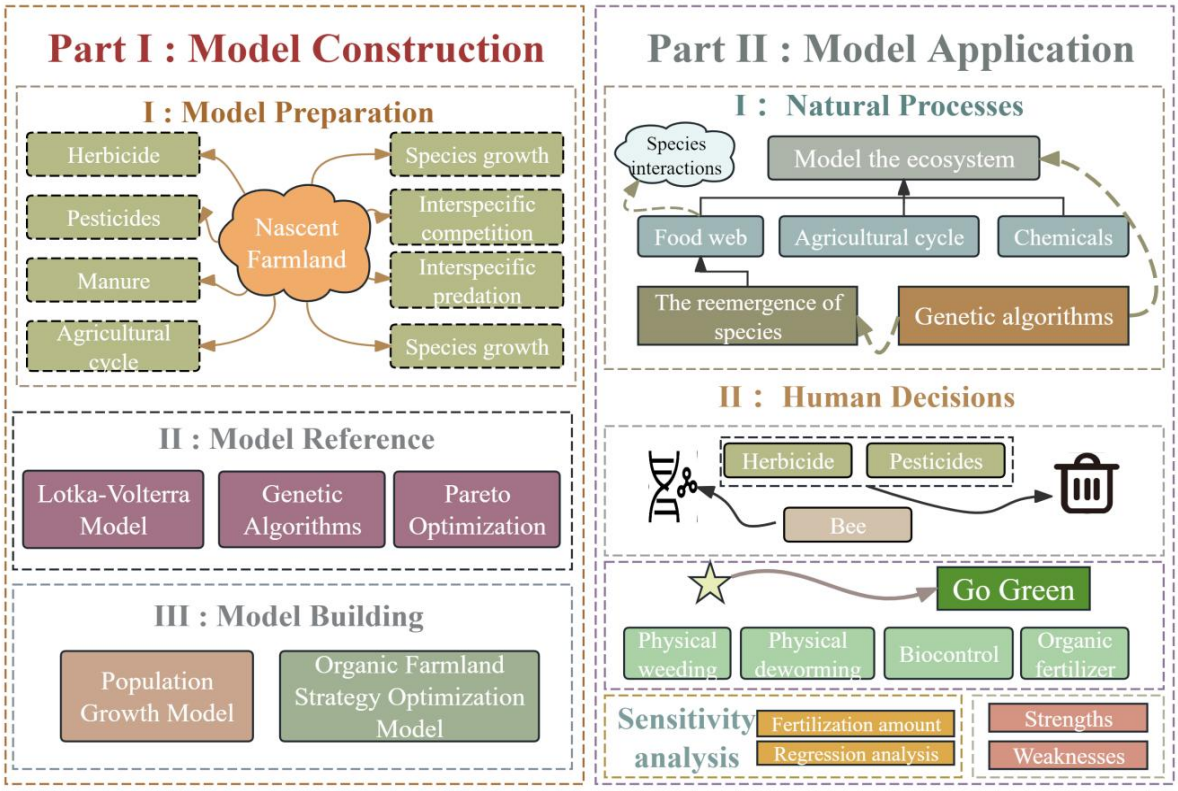


Figure 3: the Flow Chart of our work

2. Assumptions and Justifications

To simplify the problem and facilitate the simulation of real-world scenarios, we make the following fundamental assumptions, each with a reasonable basis:

- **Assumption 1: No pesticide resistance develops in the biological populations within the ecosystem.**
  Justification: Given the relatively short simulation period of the farmland ecosystem, the accumulation of pesticide resistance can be largely ignored and does not significantly affect ecosystem dynamics.
- **Assumption 2: No predation occurs within the same species, and predation is limited to specific interspecies interactions, with no reciprocal predation.**
  Justification: This assumption simplifies the model while preserving critical interspecies interactions essential for ecosystem dynamics.
- **Assumption 3: The time required for planting and harvesting crops is negligible compared to their growth cycle.**
  Justification: In most agricultural systems, planting and harvesting time constitutes a small fraction of the crop lifecycle and has minimal impact on overall system dynamics.
- **Assumption 4: Introduced pollinators (e.g., bees) do not compete with other species in the ecosystem and solely promote crop growth through pollination.**
  Justification: Pollinators are assumed to function exclusively as facilitators of crop growth, avoiding changes in existing competitive dynamics. This simplifies their roles in the ecosystem (e.g., competition for resources or space) and highlights their core function.

3. Notations

Table 1. Notations used in this paper

Symbol	Description
$\lambda_{i,j}$	The interaction coefficient between the i-th organism and the j-th organism in the food web
$\eta_{i,j}$	Energy utilization rate of species i to species j
N_i	The population size of species i
r_i	The natural growth rate of species i
K_i	Environmental carrying capacity of species i
γ_i	Natural mortality rate of species i
τ_x	The influence of external environment on crop growth

4. Model I :Population Growth Model

4.1 Data Collection

We primarily utilize the following data: information on species present before forest conversion, details on how the forest was transformed into farmland, pesticide and herbicide usage, crop rotation, and species changes. Table 1 summarizes the data sources.

Table 2. Data and Database Websites

Data Website	Data Type
https://www.globalforestwatch.org/	Biodiversity indicators
http://www.fao.org/faostat/en/	Chemical utilization
https://www.nass.usda.gov/	Agricultural production
https://www.eea.europa.eu/data-and-maps/data/land-cover-accounts-leac	Land change

4.2 Model Overview:

When constructing a new agricultural ecosystem model and evaluating the impacts of herbicides and pesticides, we choose the Logistic Growth Model [1], Lotka-Volterra equations, and Allee Effect model. The Logistic Growth Model is well-suited for describing the saturated growth of crops and weeds under limited resources, reflecting the natural constraints of environmental carrying capacity. The Lotka-Volterra model reveals how pesticides indirectly affect the population dynamics of bats and birds by reducing pest populations, while the Allee Effect model incorporating a seasonal amplitude term into it, capturing the periodic variations in population growth rates [2]. By integrating these models, a comprehensive assessment of the multifaceted impacts of chemical use on agricultural ecosystems can be achieved.

4.3 Model Construction

4.3.1 Sub-Model 1:Predation & Energy Utilization Rate Matrix

The natural growth rate of organisms is typically expressed as:

$$r_n = \sum_{i=1}^{n} \sum_{j=1}^{n} \eta_{i,j} \lambda_{i,j} \quad i \neq j \tag{1}$$

If we label the organisms in the food chain sequentially as producers, primary consumers, secondary consumers, etc., a matrix representation of the above equation can be constructed. This matrix encapsulates the interactions and dependencies among the trophic levels within the ecosystem.

Predation coefficient matrix for n species in an ecosystem:

$$\boldsymbol{\lambda} = \begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,n} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{n,1} & \lambda_{n,2} & \cdots & \lambda_{n,n} \end{bmatrix} \quad \boldsymbol{\lambda} = \begin{bmatrix} 0 & \lambda_{1,2} & \cdots & \lambda_{1,n} \\ 0 & 0 & \cdots & \lambda_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \tag{2}$$

(a) (b)

According to Hypothesis 3, due to the food chain in our farmland ecosystem, lower-level producers can not prey on higher-level consumers. Therefore, the matrix is essentially an upper triangular matrix, where $\lambda_{i,j} = 0$ when $i \geq j$, as shown in matrix (2)(b).

Energy utilization efficiency matrix for n species in an ecosystem:

$$\boldsymbol{\eta} = \begin{bmatrix} \eta_{1,1} & \eta_{1,2} & \cdots & \eta_{1,n} \\ \eta_{2,1} & \eta_{2,2} & \cdots & \eta_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \eta_{n,1} & \eta_{n,2} & \cdots & \eta_{n,n} \end{bmatrix} \quad \boldsymbol{\eta} = \begin{bmatrix} 0 & \eta_{1,2} & \cdots & \eta_{1,n} \\ 0 & 0 & \cdots & \eta_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \tag{3}$$

(a) (b)

Here, $\eta_{n,n}$ represents the energy utilization efficiency of a specie that preys on another specie. Similarly, the actual energy utilization efficiency matrix can be simplified to matrix (3)(b) in this model. We observe that the simplified expression for natural growth rates uses the Hadamard product (element-wise multiplication) of the element-wise multiplication of the corresponding entries in two matrices. Each element of the Hadamard product $(\boldsymbol{\eta} \circ \boldsymbol{\lambda})_{i,j}$ represents the total ecological contribution of one specie's predation behavior on another in the ecosystem.

Thus, the final result can be written as:

$$r_n = \sum_{i,j} (\boldsymbol{\eta} \circ \boldsymbol{\lambda})_{i,j} \quad \text{or} \quad r_n = \text{sum}(\boldsymbol{\eta} \circ \boldsymbol{\lambda}) \tag{4}$$

4.3.2 Sub-Model 2: Logistic Growth Model for Crop Growth.

In the model, the rate of change of crop populations is represented by the following equation [1]:

$$\frac{dN_1}{dt} = r_1 \cdot N_1 \cdot \left(1 - \frac{N_1}{K_1}\right) - \lambda_{1,3} \cdot N_1 \cdot N_3 - \lambda_{1,2} \cdot N_1 \cdot N_2 \tag{5}$$

4.3.3 Sub-Model 3: Introducing the Allee effect for the seasonal amplitude term

It is taken into account in the expression for the crop growth rate, in particular the introduction of the seasonal amplitude term, which is manifested in the growth rate r_1 parameterization [3]:

$$r_1 = \left(r_{1,0} + \tau_F \cdot F \cdot \exp(-\beta_F \cdot (t - t_i))\right) \cdot \left(1 + A \cdot \sin\left(\frac{2\pi t}{T_s}\right)\right) \tag{6}$$

where $r_{1,0}$ denotes the basic growth rate of the crop, τ denotes the growth regulation coefficient, which indicates the effect of external influences on the growth of crops, F is the amount of fertilizer applied to the field, β is the decay rate of fertilization efficacy, t_i is the starting time for fertilization, A is the seasonal amplitude of crop growth, T_s is the seasonal cycle of the crop.

Parameterization of population carrying capacity:

$$K_1 = K_{1,0} \cdot \left(\exp(-\iota \cdot t) + \lambda \cdot (T_s - 1)\right) + \lambda \cdot F \cdot \exp(-\beta_F \cdot (t - t_i)) \tag{7}$$

where K_1 denotes the carrying capacity of the crop population, $K_{1.0}$ denotes the initial carrying capacity of the crop population, and similarly for other species, T is the period of the cycle.

Weed population equations:

$$\frac{dN_2}{dt} = r_2 \cdot N_2 \cdot \left(1 - \frac{N_2}{K_2}\right) - v_1(t) \cdot V_1 - \lambda_{1,2} \cdot N_1 \cdot N_2 \tag{8}$$

$$v_1(t) = V_{1,0} \cdot \exp\left(-\phi_1 \cdot \left(t - t_i\right)\right) \tag{9}$$

where $v_1(t)$ is the efficacy of the herbicide, $V_{1,0}$ is the basic efficacy of the herbicide; $v_2(t)$ is the efficacy of the pesticide, $V_{2,0}$ is the basic efficacy of the pesticide, V_1 is the amount of herbicide, V_2 is the amount of pesticide, ϕ_1 is the decay factor of the chemical, representing the decline in efficacy over time.

Pest population equations:

$$\frac{dN_3}{dt} = r_3 \cdot N_3 \cdot \left(1 - \frac{N_3}{K_3}\right) - v_2(t) \cdot V_2 + \lambda_{1,3} \cdot N_1 \cdot N_3 - \lambda_{3,4} \cdot N_4 \cdot N_3 \tag{10}$$

$$v_2(t) = V_{2,0} \cdot \exp\left(-\phi_2 \cdot \left(t - t_i\right)\right) \tag{11}$$

$$K_3(t) = K_{3,0} + \rho_{N_1} \cdot N_1 + \rho_{N_2} \cdot N_2 \tag{12}$$

$$\rho_{N_1} + \rho_{N_2} = 1 \tag{13}$$

where ρ denotes the environmental impact factor. Indicates, ρ_{N_i} is the amount of environmental capacity provided by crops for pests.

Bat population equations:

$$\frac{dN_4}{dt} = \lambda_{3,4} \cdot N_3 \cdot N_4 - \gamma_4 \cdot N_4 \tag{14}$$

where γ_4 denotes the natural mortality of bats.

Seeding equation:

$$\begin{cases} N_1(0^+) & = N_{1,0} \\ N_1(T_s^-) & = N_1(t) \\ N_1(T_s) & = 0 \\ N_1(T_s^+) & = N_{1,0} \end{cases} \tag{15}$$

where T_s^- indicates the number of crops at the moment before harvesting, and T_s^+ denotes the number of crops at the moment after harvesting. This process indicates that crops are planted again in a shorter period of time after harvesting, and for simplicity of the model the time of planting the crops is ignored.

Simpson's index and Ecosystem stability:

$$D = 1 - \sum_{i=1}^{n} \left(\frac{N_i}{N}\right)^2 \tag{16}$$

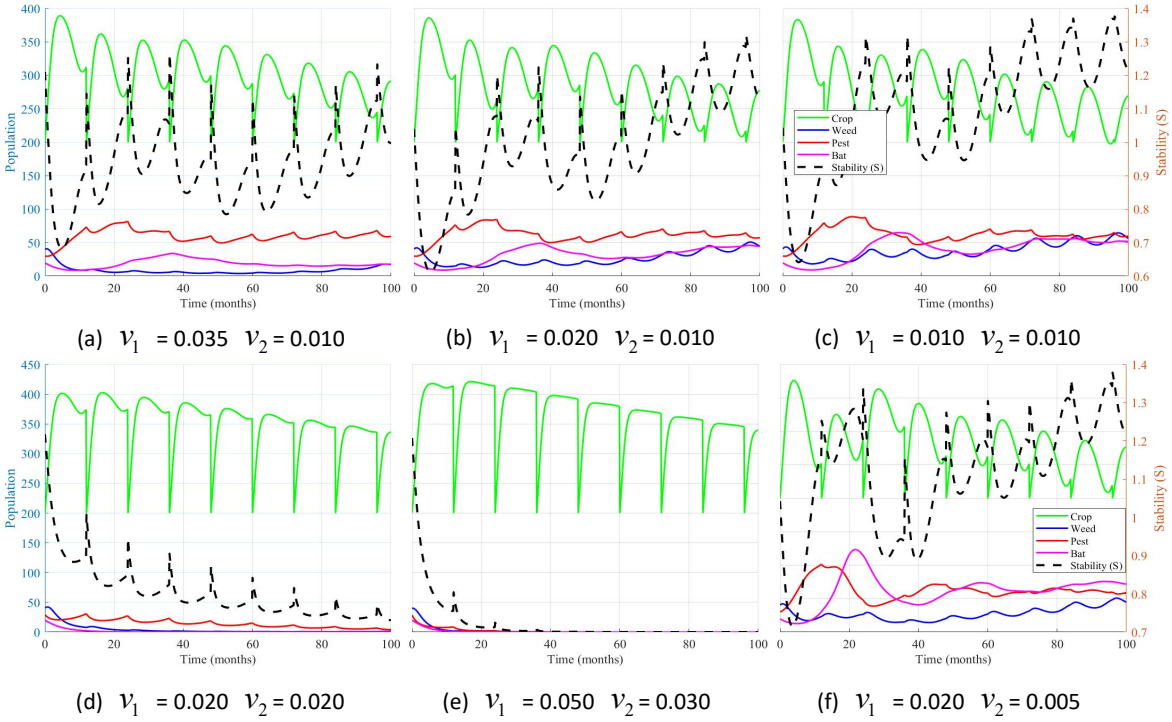
$$S = D \cdot \ln(1 + \alpha_F F) \cdot e^{-\alpha_{V_1} \cdot V_1} \cdot e^{-\alpha_{V_2} \cdot V_2} \tag{17}$$

$$\alpha_F + \alpha_{V_1} + \alpha_{V_2} = 1 \tag{18}$$

where α_F denotes the weight of the impact of the amount of fertilizer applied to the farmland on the farmland ecosystems [4].

These models are combined to construct a foundational food web model for simulating a new agricultural ecosystem. The model incorporates producers, consumers, seasonal variations in the agricultural cycle, and the effects of chemical usage on plants, insects, bats, and overall ecosystem stability.

4.4 The impacts of chemical use on ecosystem stability



(a) $v_1 = 0.035$ $v_2 = 0.010$ (b) $v_1 = 0.020$ $v_2 = 0.010$ (c) $v_1 = 0.010$ $v_2 = 0.010$

(d) $v_1 = 0.020$ $v_2 = 0.020$ (e) $v_1 = 0.050$ $v_2 = 0.030$ (f) $v_1 = 0.020$ $v_2 = 0.005$

Figure 4: Farmland Eco system Simulation

We first use data collected from the literature,after preprocessing, conduct computer simulations and generated Figure 4. Subsequently, data are extracted from the time-series plots, normalized, and then organized into a table for the three core indicators (crop yield, ecosystem stability, and biodiversity) under six scenarios: low chemical use, moderate chemical use, high herbicide use, high pesticide use, excessive chemical use, and insufficient chemical use. A heatmap is then created to visualize these results. Here, "chemicals" represent a combination of herbicides and pesticides, used to conveniently illustrate their shared trends. The heatmap effectively visualizes the performance of each indicator under different scenarios, making it easier to compare and analyze the impact of various chemical use strategies on the agricultural ecosystem.

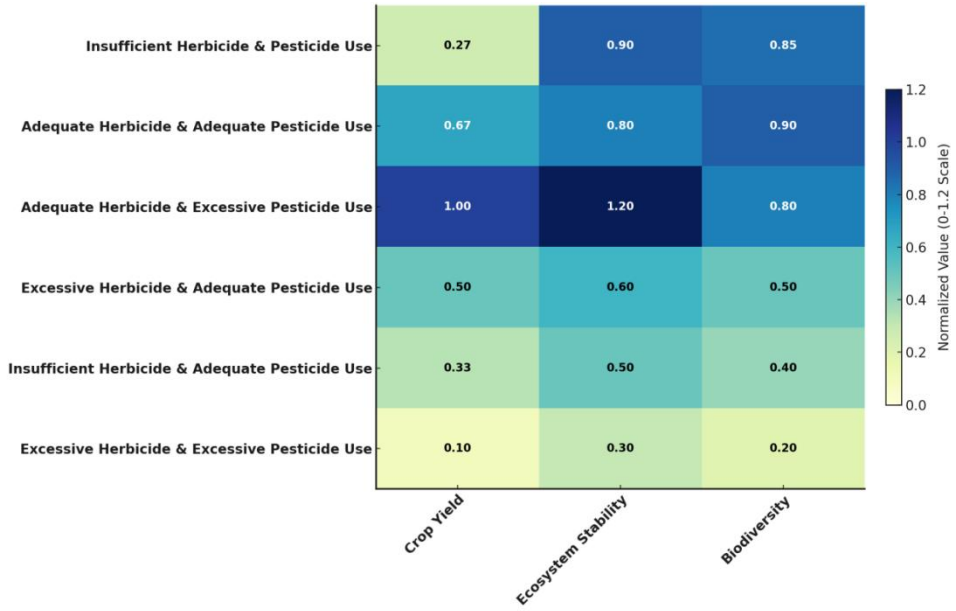


Figure 5: Heat Map of Herbicide and Pesticide Use Impact on Ecosystem and Crop Yield

Smaller numbers correspond to lower impacts in the heat map. From the visualization results in Figure 5, we observe that under low pesticide application intensity, crops are significantly affected by pests and weeds, leading to a noticeable decline in yield over time. However, ecosystem diversity remains high while stability remains strong. In contrast, when pesticide application intensity is excessively high, the destructive effects of chemicals on the farmland ecosystem become pronounced. Although target species such as weeds and pests are well suppressed, the populations of predators like bats decline significantly, and crop yield shows little improvement or even decreases. These findings indicate that the choice of chemical application levels significantly affects crop yield, target species (weeds and pests), and ecosystem stability.

5. Model II：Optimal Chemical Usage Model

5.1 Model Overview

In the second task, we build upon the study by Simmons, N. B., to clarify the species types identified in the first task. We use corn as the primary crop in the farmland ecosystem, represented by green lines, and introduce weeds as competitors to corn, represented by blue lines. In addition to a pest for corn :the corn borer, we introduce cotton bollworm, which competes with the corn borer. Eptesicus serotinus is included as the bat specie preying on pests. Considering the gradual introduction of native species into the farmland ecosystem over time, Falco peregrinus is added as a bird specie serving as the top predator, preying on bats. The model from the first task is improved and optimized using a genetic algorithm to obtain the results.

5.2 Ecological model of introduced new species

The following are the agricultural ecosystem model equations after adding the new species Helicoverpa armigera and peregrine falcon. These extensions simulate more complex ecological interactions while evaluating the impact of chemical use and ecological succession on system stability.

The corn population equation optimized based on Equation (5):

$$\frac{dN_1}{dt} = r_1 \cdot N_1 \cdot \left(1 - \frac{N_1}{K_1}\right) - \lambda_{1,3} \cdot N_1 \cdot N_3 - \lambda_{1,2} \cdot N_1 \cdot N_2 - \lambda_{1,5} \cdot N_1 \cdot N_5 \tag{19}$$

The corn borer population equation optimized based on Equation (10):

$$\frac{dN_3}{dt} = r_3 \cdot N_3 \cdot \left(1 - \frac{N_3}{K_3}\right) + \lambda_{1,3} \cdot N_1 \cdot N_3 - \lambda_{3,5} \cdot N_3 \cdot N_5 - \lambda_{3,4} \cdot N_3 \cdot N_4 - v_2(t) \cdot V_2 \tag{20}$$

The cotton bollworm population equation:

$$\frac{dN_5}{dt} = r_5 \cdot N_5 \cdot \left(1 - \frac{N_5}{K_5}\right) + \lambda_{1,5} \cdot N_1 \cdot N_5 - \lambda_{3,5} \cdot N_3 \cdot N_5 - v_2(t) \cdot V_2 \tag{21}$$

The Eptesicus serotinus population equation optimized based on Equation (14).

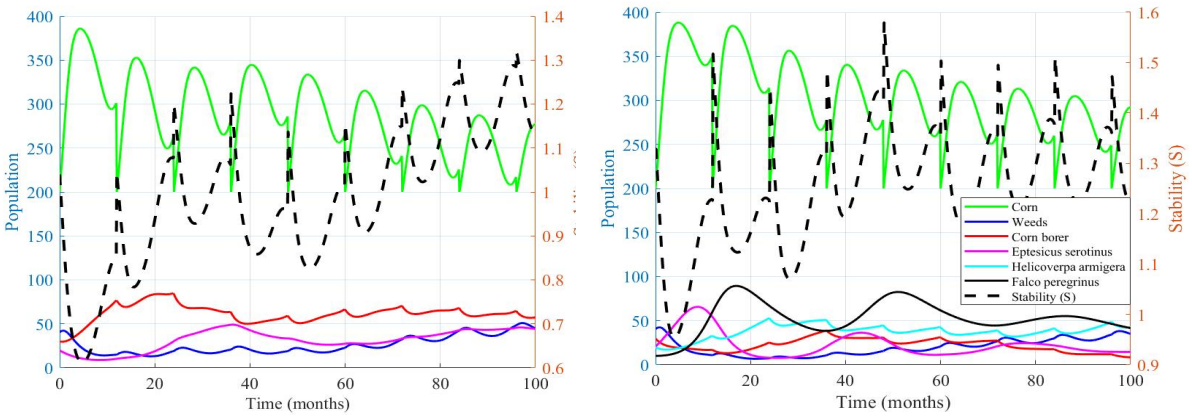
$$\frac{dN_4}{dt} = r_4 \cdot N_4 \cdot \left(1 - \frac{N_4}{K_4}\right) + \lambda_{3,4} \cdot N_3 \cdot N_4 - \lambda_{4,6} \cdot N_4 \cdot N_6 \tag{22}$$

The Peregrine falcon population equation.

$$\frac{dN_6}{dt} = \lambda_{4,6} \cdot N_4 \cdot N_4 - \gamma_6 \cdot N_6 \tag{23}$$

Model II extends Model I and use the Genetic Algorithm to simulate and optimize the interactions within agricultural ecosystems. This model introduces species reemerging during ecological succession, such as the Peregrine falcon and Cotton bollworm, and incorporates the temporal decay effects of chemical use (e.g., fertilizers, herbicides, and insecticides). Ecosystem stability is quantified using an ecological stability index.

By employing the Genetic Algorithm, the model optimizes chemical use strategies to balance ecological stability, economic returns, and operational costs. Model II also captures dynamic interactions among species, accounts for seasonal variations, and evaluates the long-term impacts of agricultural practices and species recovery. This provides a robust tool for achieving sustainable agriculture and ecosystem conservation. Substituting parameter values from the literature yields the to form graphical results Figure 6:



(a)Befor (a)After

Figure 6: Changes due to the emergence of native

Figure (a): The ecosystem is relatively simple, primarily consisting of corns, pests, and weeds. Pest predation and periodic corn planting serve as the primary driving factors.

Figure (b): The introduction of Cotton bollworm (cyan line) adds new predation and competition dynamics to the system. The inclusion of Peregrine falcon (black line) exerts significant pressure on the bat population, causing more pronounced fluctuations, though maintains a stable population.

5.3 Overall Comparison

Table 3. Before and after introduction comparative Analysis

Feature	Before Introduction	After Introduction
Population Dynamics	Corn and pest populations highly unstable	Corn more stable, pests controlled effectively
Stability Index S	Lower (1.0–1.3)	Higher (1.0–1.6), indicating better stability
Food Web Structure	Simple, only corn, pests, and weeds	Complex, with added predation and competition
Pest Control	Pests exert strong pressure on corn	Bats control pests effectively, reducing pressure

As shown in Figure 6 and compared in Table 3, the introduction of new species significantly increases the complexity and stability of the ecosystem. Before species introduction, the ecosystem is relatively simple, with notable fluctuations in the populations of corn, weeds, and pests. Pests exert considerable pressure on corn, resulting in low ecosystem stability. After the introduction of Peregrine falcons, pests are effectively controlled, reducing pest pressure and stabilizing the corn population.

5.4 Introducing Genetic Algorithm optimization

To quantitatively study the impact of introducing two different species on the ecosystem, we use a genetic algorithm to perform iterative analysis on the farmland ecosystem model based on Lotka-Volterra equations from the Task 1 and the modified model with newly introduced species from the Task 2. Our goal is to identify the optimal solution [5].

5.4.1 Multi-Objective Weighted Optimization Fitness Function

A multi-objective weighted optimization utility function is a formula designed to balance multiple objectives, such as ecological stability, economic benefits, and operational costs. Adjustable weight parameters are used to assign relative importance to each objective.

$$\begin{cases} f(F,V_1,V_2) & = \omega_S \cdot \ln(1+S) + \omega_I \cdot \ln(1+\text{Income}) - \omega_T \cdot \ln(1+\text{Total_cost}) \\ 1 & = \omega_S + \omega_I + \omega_T \\ \text{Total_cost} & = F \cdot c_F + V_1 \cdot c_{V1} + V_2 \cdot c_{V2} \\ \text{Income} & = \overline{Y} \cdot c_{crop} \\ \overline{Y} & = \frac{1}{T}\int_0^T N_1(t)dt \end{cases} \tag{24}$$

5.4.2 Selection Probability Formula

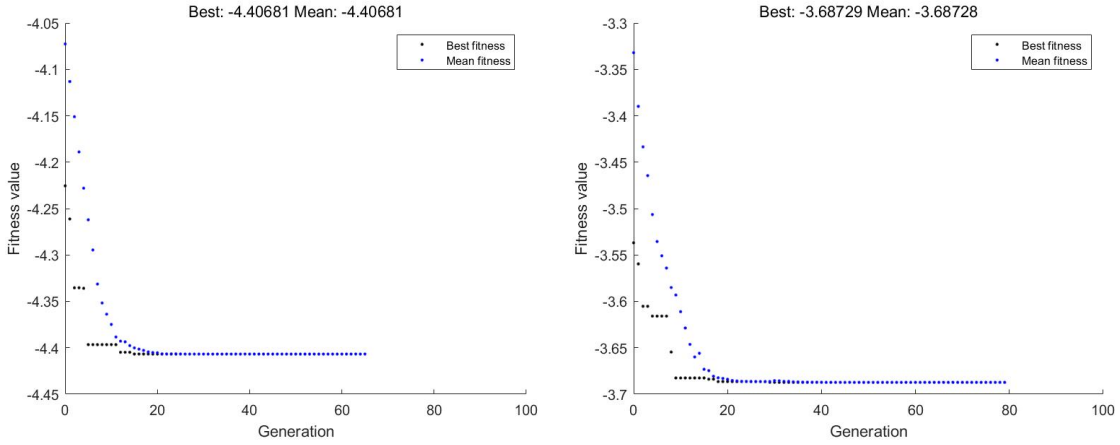
We incorporate the nonlinear weighting effects of the fitness function and enhance the selection process by assigning the probability of individual selection proportional to their fitness values relative to the total fitness of the population. This refinement improves its efficiency and convergence towards optimal solutions.

$$P_i = \frac{f_i}{\sum_{j=1}^{N_j} f_j} \tag{25}$$

The crossover formulas and mutation formulas in the genetic algorithm are consistent with those in the classical one and remain unchanged.

5.4.3 Optimization results

In the Task 2, the iterative process of the genetic algorithm is shown in Figure 7. Stable results are achieved after 60–80 iterations. Through this progress, it is observed that the introduction of new species in Task 2 increases the complexity of the ecosystem. The model in Task 2 demonstrates a more significant improvement in fitness value (-3.68729 > -4.40681), indicating that the balance among ecological stability, economic returns, and costs is better optimized.



(a)Befor (b)After

Figure 7: Genetic algorithm optimization before and after species

Additionally, we quantify the impact of species reintroduction on the agricultural ecosystem by applying genetic algorithm optimization to Model I and Model II. A comparison of the data yields Table 4:

Table 4. Comparison of Genetic Algorithm results

	Before species return	After species return
Optimal F	5.0000	5.0000
Optimal V1	0.0092	0.0010
Optimal V2	0.0313	0.0265
Stability S	0.0007	0.0037
Average crop yield	354.3522	377.8184
Total cost	11.7160	11.1097
Comprehensive benefit	3.0154	3.0642

The introduction of Cotton bollworm and Peregrine falcon significantly enhances the stability of farmland ecosystem, as demonstrated by the following:

a) Improved ecosystem stability: The stability coefficient increases by 428%, indicating a substantial enhancement in biodiversity and ecosystem resilience.

b) Increased average crop yield: Crop yield rises by approximately 6.6%, with reduced fluctuations, leading to more stable production.

c) Reduced total maintenance costs: Total costs decrease by about 5.2%, reflecting a reduced dependence on chemical inputs.

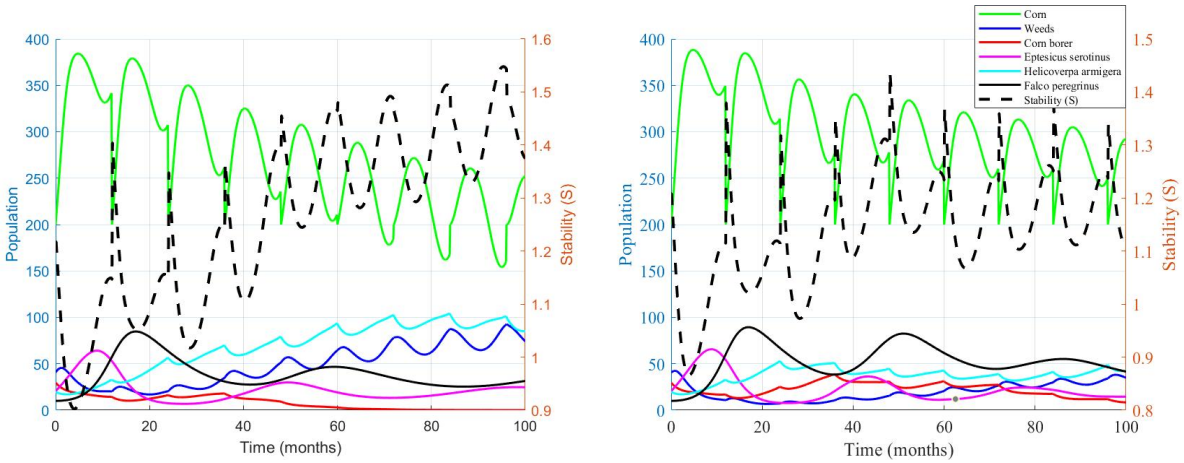
The improvement in the fitness function value indicates that the ecological management strategy based on biocontrol is more effective, balancing environmental stability, economic returns, and crop yield, ultimately enhancing the comprehensive benefit of the farmland ecosystem.

6. Model III : Farmland Ecological Regulation Model

As the ecosystem gradually matures, farmers remove herbicides and analyze the stability of the ecosystem before and after it's removal from producers and consumers. Following the removal of herbicides, the dynamic balance between producers and consumers faces significant challenges. To address these changes, bees are introduced as an additional ecological regulator species. The model integrates multi-species interactions, the pollination roles of bees and bats, the dynamic balance of the ecosystem, and the optimization objectives of sustainable agriculture.

6.1 Herbicide Removal

In the ecosystem prior to the introduction of bees, analyze the stability of the ecosystem before and after the herbicide removal, the results are presented in Figure 8.



(a) (b) After

Figure 8:Population growth before and after herbicide removal

Impact of Herbicide Removal:

In the absence of herbicides, weed competition intensifies significantly, leading to a decline in corn yield. Simultaneously, the suppression of pests such as the corn borer and cotton bollworm weakens, resulting in an increase in pest populations. This causes a short-term decline in ecosystem stability. However, over time, the ecosystem gradually achieves self-regulation in the absence of chemical interventions, leading to long-term stability.Bats play a critical role in this process as insectivores, through their interactions with insects and predators, establishing a dynamic balance, enhancing the ecosystem's resilience.

6.2 The Ecological Role of Bees

Pollination Role of Bees:

The primary role of bees in the ecosystem is pollination, which indirectly influences ecological dynamics by altering plant reproduction and growth patterns. Bees can pollinate corn, the model parameters are configured to avoid conflicts.

Whether Bees Compete with Other Organisms:

In the current ecosystem, the primary role of bees is pollination, and do not directly compete with existing species such as bats or pests:

Bees vs. Bats: The activity ranges and times of bees and bats are distinct. Bees are active

during the day, while bats are nocturnal, occupying different ecological niches.

Bees vs. Pests: Bees neither prey on nor compete for plant resources, resulting in minimal interaction with pests.

Bees vs. Peregrine Falcons: Peregrine falcons prey on bats, but bees are not part of their diet.

6.3 The Impact of bees on the corn population equation

Bats are added to the food web as pollinators. The food web of this ecosystem is now as shown in Figure 9.

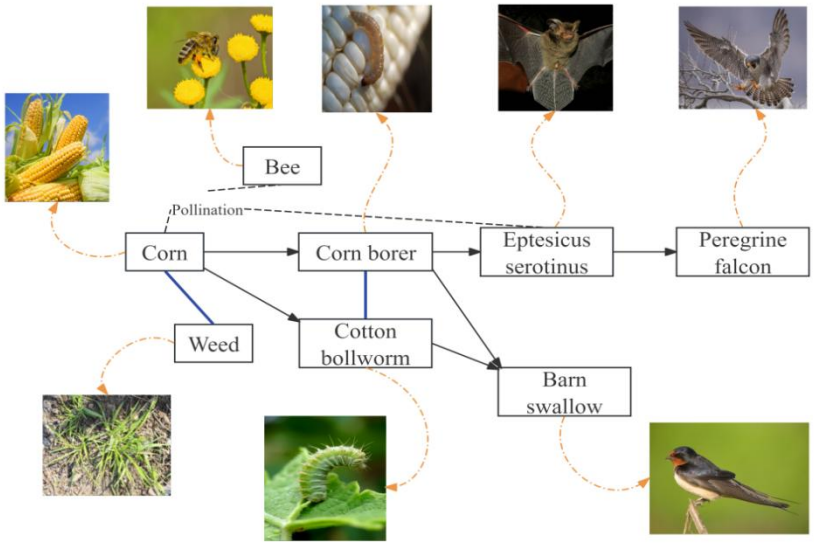


Figure 9：Food weds and correspondence

The corn population change rate equation optimized based on equation (6):：

$$r_1 = \left(r_{1,0} + \tau_F \cdot F \cdot e^{-\beta_F \cdot (t-t_i)} + \tau_{N_4} \cdot F \cdot e^{-\beta_{N_4} \cdot (t-t_i)} + \tau_{N_7} \cdot Fe^{-\beta_{N_7} \cdot (t-t_i)}\right) \cdot \left(1 + A \cdot \sin\left(\frac{2\pi t}{T_s}\right)\right) \quad (26)$$

where $r_{1,0}$ denotes the baseline growth rate of corn, τ_F denotes the growth regulation coefficient of fertilizer, τ_{N_4} denotes the growth regulation coefficient of pollination by bats, τ_{N_7} denotes the growth regulation coefficient of pollination by bees, β_F denotes the decay rate of fertilizer effectiveness, β_{N_4} denotes the decay rate of pollination effectiveness by bats [6].

The bee population equation:

Bees play a crucial role in pollination, and their population dynamics directly influence crop production efficiency and ecosystem balance.

$$\frac{dN_7}{dt} = r_7 \cdot N_7 \cdot \left(1 - \frac{N_7}{K_7}\right) \quad (27)$$

where N_7 denotes the bee population size, r_7 denotes the growth rate of the bee population.

This set of equations describes the complex interactions among corn, bees, bats, pests, and competitors within the agricultural ecosystem. Through modeling, it enables the analysis of how the interactions between producers and consumers influence the overall stability of the ecosystem in the absence of chemical inputs.

Under the condition of herbicide removal, the introduction of bees is shown in Figure 10. By comparing this with Figure 8(b), a radar chart (Figure 11) is drawn to visually compare

the impact of bees.

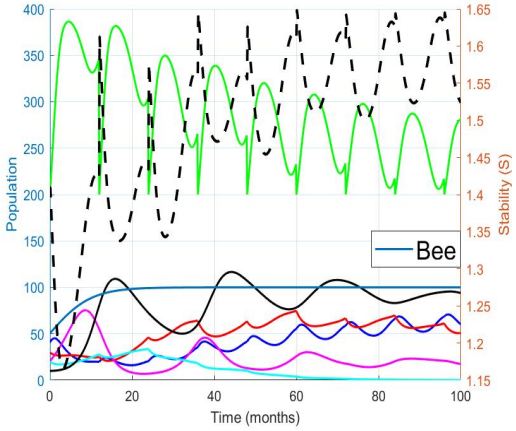


Figure 10：Population Growth with

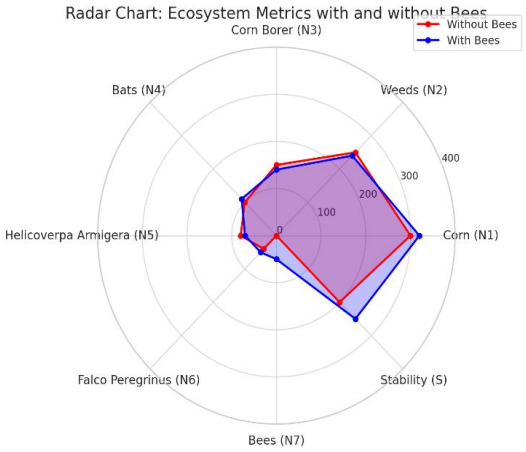


Figure 11: Comparison before and after the introduction

The introduction of bees has a significant impact on the ecosystem in various ways. In the radar chart, the red line represents the ecosystem state without bees, while the blue line represents the state with bees. Each axis represents an ecological indicator, such as "Corn" for crop yield, "Weeds" for weed population, and "Stability" for ecosystem stability.

The introduction of bees brings the following benefits: Bees significantly increase corn yield through pollination. Populations of corn borers and cotton bollworms decrease significantly after it. It also improves the stability index, making the system more balanced and diverse. Furthermore, the introduction of bees indirectly reduces the number of weeds, possibly due to the increased crop coverage area inhibiting weed growth.

In conclusion, the introduction of bees provides significant benefits to producers and the health of the ecosystem and plays a crucial role in maintaining ecological balance.

6.4 Genetic algorithm of Model III

The principles and formulas of the genetic algorithm are detailed in Section 5.4. Here, we present a bar chart for the genetic algorithm applied to Model III, quantitatively illustrating the impact of introducing bees on the ecosystem, as shown in Figure 12:



	Optimal F	Optimal V2	Stability S	Average crop yield	Total cost	Comprehensive benefit
Before	5	0.0202	0.0044	346.9684	10.8093	3.0184
After	5	0.0188	0.0073	354.3383	10.7527	3.0325

Figure 12 ： The optimal solution before and after the introduction of

The analysis of Figure 12 shows that the introduction of bees has a significant positive impact on the ecosystem: crop yield increases from 346.97 to 354.34, enhancing the economic benefits for producers; pest populations are suppressed, reducing pesticide usage from 0.0202 to 0.0188, thereby decreasing dependence on chemical measures; and ecosystem stability improves notably, rising from 0.0044 to 0.0073.

This demonstrates that the introduction of bees is a sustainable strategy that balances both ecological and economic benefits.

7. OFC Decision Optimization Model

7.1 Model Overview

With the growing emphasis on environmental sustainability, organic agriculture is regarded as a critical pathway for achieving sustainable agricultural transformation. In response, we establish an organic farmland combination decision optimization model, including the following components:

A) Pareto Frontier Analysis: We perform Pareto frontier analysis based on Model III to examine the dynamic interactions among Farmland Status, Ecosystem Stability, and Cost. Multi-objective optimization is employed to reveal the trade-offs between these factors and to identify the Pareto optimal solutions.

B) RGB Ecosystem Cellular Automata: Using a discrete grid and the Moore neighborhood method, we simulate the spatial dynamics of the farmland ecosystem under different strategy combinations. RGB visualization is employed to present the spatial distribution of farmland cost-effectiveness, ecosystem stability, and crop health.

7.2 Pareto Optimization Model

In this problem, the objective functions are defined as follows:

$$f(x) = \left(f_1(x), f_2(x), f_3(x)\right) = \left(R_C(t), R_G(t), R_D(t)\right) \tag{29}$$

Cost-Effectiveness Function R_C(t) :

$$R_C(t) = t_{OFC} \cdot \left(\frac{c_{OFU} \cdot B_{OFU}}{T_{OFU}} + \left(c_{N_8} + \frac{c_{BPC}}{T_{BPC}} \right) \cdot B_{BPC} + \frac{c_{PWC}}{T_{PWC}} \cdot B_{PWC} + \frac{c_{PPC}}{T_{PPC}} \cdot B_{PPC} \right) \tag{30}$$

$$\mathrm{B} = \text{Pareto-Bool} = \begin{cases} 1, & \text{use i,} \\ 0, & \text{no i,} \end{cases} \quad \text{where } i \in \{\text{OFU,PWC,PPC,BPC}\}. \tag{31}$$

Here, OFC represents Organic Farming Components, OFU stands for Organic Fertilizer Use, PWC refers to Physical Weed Control, PPC denotes Physical Pest Control, and BPC represents Biological Pest Control.

Farmland Status Function R_G(t):

$$R_G(t) = \begin{cases} N_1 - (N_3 + N_5), & \text{if } (N_3 + N_5) > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{32}$$

Ecosystem Stability Function R_D(t)：

$$N_{\text{total}} = \sum_{i=1}^{8} N_i \tag{33}$$

$$R_D(t) = \begin{cases} 1 - \sum_{i=1}^{8} \left(\frac{N_i}{N_{\text{total}}} \right)^2, & \text{if } N_{\text{total}} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (34)$$

These three objective functions reflect the balance between yield, stability, and cost within the agricultural ecosystem.

Figure 13 illustrates the organic decision-making process developed in this study.

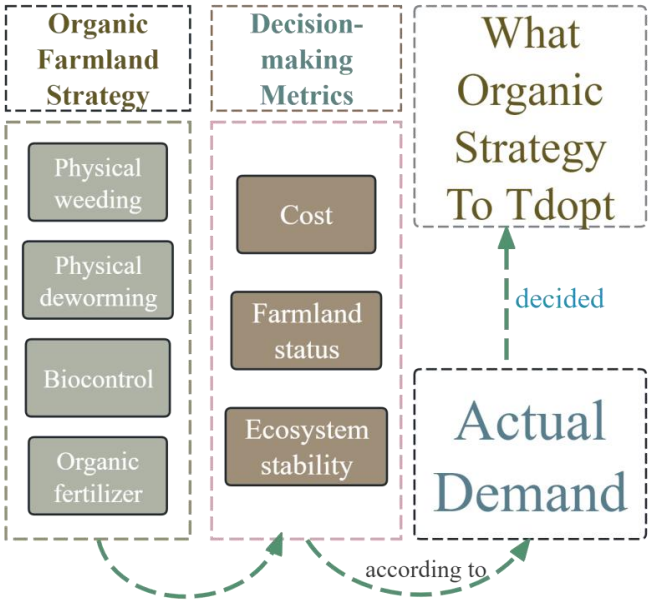


Figure 13: Organic Decision making based on Pareto Frontier

7.3 Pareto Frontier Solution for Organic Agriculture Decisions

The table 5 presents the results of the Pareto Frontier Solution. It demonstrates the outcomes for Farmland Status, Ecosystem Stability, and Cost under different combinations of organic fertilizer use, physical weeding, physical pest control, and biocontrol strategies.

Table 5. Pareto Frontier Solutions for Organic Farming Decision Making

Organic Fertilizer	Physical Weeding	Physical Deworming	Biocontrol	Farmland Status	Ecosystem Stability	Cost	Scenario
0	0	1	0	380.9	0.31683	200	1
0	1	0	0	-23.653	0.64342	83.333	2
0	1	1	0	386.82	0.30104	283.33	3
0	0	0	1	132.47	0.76093	266.67	4
0	0	1	1	380.9	0.41138	466.67	5
0	1	0	1	167.66	0.68062	350	6
0	1	1	1	386.82	0.3967	550	7
1	0	1	0	383.63	0.32846	216.67	8
1	1	0	0	-25.581	0.65843	100	9
1	1	1	0	387.45	0.3153	300	10
1	0	0	1	154.87	0.77752	283.33	11
1	0	1	1	383.63	0.42778	483.33	12
1	1	0	1	178.3	0.70786	366.67	13
1	1	1	1	387.45	0.41561	566.67	14

Based on the above Pareto Frontier Solutions, we can analyze the advantages and disadvantages of the agricultural components and use these insights to inform agricultural strategies from the following perspectives:

a) Pest Control: Physical pest control (Scenario 1) and biocontrol (Scenario 4) are effective methods for reducing pest threats to crops. Physical pest control is cost-effective and shows immediate results but may have negative impacts on non-target species. Biocontrol, while less effective in pest reduction, is more beneficial for ecosystem stability (0.76093).

b) Crop Health and Plant Reproduction: Pareto frontier analysis indicates that physical pest control significantly contributes to farmland status (e.g., Scenario 1), highlighting its importance for maintaining farmland health. When combined with organic fertilizers and other measures (e.g., Scenario 12), crop health improves further, but at a higher cost.

c) Biodiversity and Long-term Sustainability: Biocontrol helps restore biodiversity and promotes long-term sustainability (e.g., Scenario 4 and Scenario 11). Sole reliance on physical methods (e.g., Scenario 1), while improving farmland status, is less favorable for the long-term stability and biodiversity of the ecosystem.

d) Cost Effectiveness: Over-reliance on multi-measure combinations (e.g., Scenario 14) incurs high costs (566.67) with limited marginal ecological benefits. In contrast, moderate combinations (e.g., Scenario 5, cost 466.67) achieve a good balance between ecosystem stability (0.41138) and farmland status (380.9).

The following 3D Pareto Front Visualization illustrates the distribution of these frontier solutions.

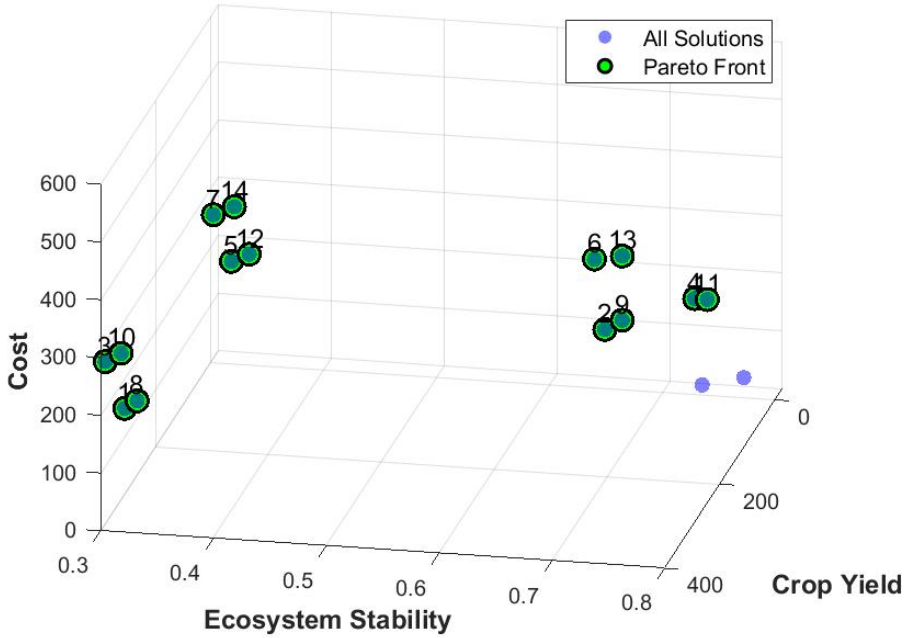


Figure 14: 3D Pareto Front Visualization

7.4 RGB Ecosystem Cellular Automata

We visualize the spatial and temporal characteristics of the farmland ecosystem by introducing cellular automata. The farmland is discretized into a $n \times n$ grid, where each cell corresponds to a local ecological unit containing components such as crops, weeds, pests, and predators. The Moore neighborhood is employed to simulate dynamic processes such as species diffusion, competition, predation, and pollination, with evolution occurring through

discrete time steps.

RGB visualization is then used to intuitively display the spatial distribution effects of different strategies on cost-effectiveness, farmland health, and ecosystem stability. This approach aids in understanding how local interactions accumulate to influence the overall ecosystem. The specific pseudocode is as follows:

Table 6: RGB Ecosystem Cellular Automaton pseudocode based on Model III

Algorithm: Predator-Equation + CA for RGB Ecosystem
1: Initialize: define an n×n grid with populations N1..N8 and predator-prey parameters
2: for step = 1 to timeSteps do
3: Compute neighborhood averages (Moore) for each Ni
4: Calculate dNi_dt based on predator-prey and competition formulas
5: Ni ← max(Ni + dNi_dt × dt, 0)
6: Periodically harvest N1 and reduce N2, N3, N5 according to farming operations
7: Compute M1, M2, M3, then normalize for R/G/B channels
8: end

Using a RGB Composite Visualization, multiple indicators' dynamic distributions can be observed from the same perspective. Researchers can quickly identify problematic areas or observe overall development trends through color changes, as shown in Figure 15.

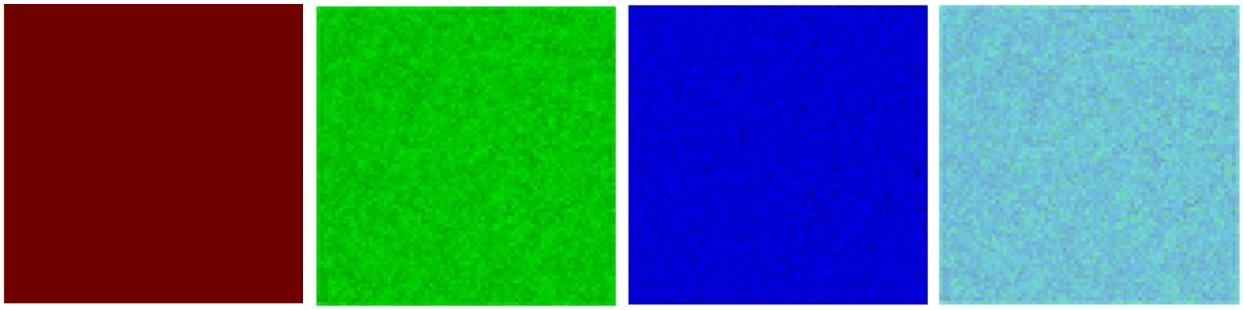


Figure 15: The operating principle of RGB Ecosystem CA

Red (Operational Cost): Highlights regions with elevated operational costs. Higher red intensity corresponds to increased resource expenditure.

Green (Farmland Status): Indicates the equilibrium between crop productivity and pest pressure. Brighter green reflects healthier and more productive farmland.

Blue (Ecosystem Diversity): Represents ecological stability and species richness. Higher blue intensity signifies greater biodiversity and a more resilient ecosystem.

This visualization framework enables simultaneous assessment of key metrics, allowing researchers to identify critical regions or observe macro-level trends in ecosystem development. Figure 15 exemplifies this approach, showcasing the dynamic evolution of common organic farming strategies over a 42-month simulation.

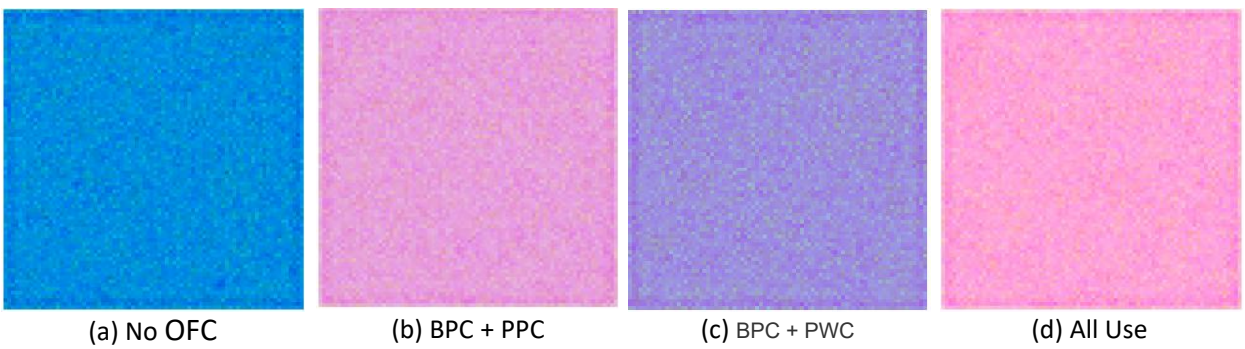


Figure 16: RGB Ecosystem CA Images using different OFCs

The visualization reveals that various strategy combinations lead to distinct outcomes, which are represented by differing colors during the cellular automaton simulation. The analysis of Figure 16 is outlined below:

(a) No OFC: Absence of Organic Farming Components: The dominant color is light cyan with a bluish tint, representing greater ecosystem diversity and stability (blue), lower farmland health (green), and reduced operational costs (red). This indicates that without intervention, the ecosystem remains stable, but the farmland health index is poor, resulting in lower crop yields.

(b) BPC + PPC: Use of Biological Pest Control and Physical Pest Control: The predominant color is light pink with a slight purple hue, reflecting higher ecosystem stability (blue), moderate farmland health (green), and average operational costs (red). This suggests that combining biocontrol and physical pest control supports ecosystem stability, although improvements to farmland health are moderate.

(c) BPC + PWC: Use of Biological Pest Control and Physical Weed Control: The color shifts toward purple with a bluish tint, indicating higher ecosystem stability (blue), moderate farmland health (green), and average operational costs (red). The combination of biocontrol and physical weed control offers minimal improvement to farmland health.

(d) All Use: Implementation of All Four Organic Farming Components: The dominant color is pink, signaling high operational costs (red), better crop yields (green), but lower ecosystem diversity (blue). This demonstrates that using all available strategies enhances crop-related outcomes and farmland health, though at the expense of increased costs and reduced ecosystem diversity.

7.5 Conclusion and Strategy Analysis

By integrating Pareto frontier analysis and RGB Ecosystem Cellular Automata, we derive several common scenarios and applicable strategies to assist farmers in decision-making:

Short-Term Low-Cost Priority Scenario: Adopting physical weeding and physical pest control is recommended. This approach is cost-effective, making it suitable for resource-constrained farmers. However, ecosystem stability is limited, making it a transitional strategy.

Ecosystem Stability Priority Scenario: Utilizing biocontrol and physical weeding is ideal. This combination maximizes ecosystem stability, promoting long-term sustainability.

Crop Yield Priority Scenario: Employing organic fertilizers and biocontrol is recommended. This strategy enhances farmland status and is suitable for farmers targeting high yields, though it involves higher costs.

Comprehensive Balance Scenario: A combination of organic fertilizers, biocontrol, and physical weeding is ideal. This approach balances ecological stability, yield, and cost-effectiveness, making it suitable for long-term sustainable development.

Specific strategies should be further optimized based on local policy support and implementation conditions.

7.6 Letter

Dear fellow farmers:

Greetings!

I am a member of the Consideration of Mature Agricultural Practices team, dedicated to researching the impact of organic farming practices on ecosystems and economic benefits. In recent years, farmers worldwide have increasingly explored how to boost crop yields while achieving sustainable agricultural development. Based on our analysis and modeling of how ecosystems evolve with various agricultural choices, we hope to provide you with practical suggestions to support your successful transition to green farming practices.

We have reviewed global data related to organic farming, including biodiversity, soil health, crop yield and quality, pest and disease control, and environmental impact. Through model calculations, we concluded that agricultural ecosystem strategies must balance ecological stability, crop yield, and cost-effectiveness.

Regarding the determination of practical strategies, our research results show that introducing multiple species such as bees and bats into the agricultural ecosystem can effectively control pests (reducing bollworm populations by 38.7%) while significantly enhancing ecosystem stability (an increase of 428%). At the same time, managing the use of chemicals such as herbicides and pesticides can prevent damage to ecosystem biodiversity and stability. Moderate use can reduce pest populations by 73.1%. Biological control has shown excellent performance in pest management and crop yield improvement. For example, the introduction of swallows increased crop yields by 32.7% and improved biodiversity by 14.2%.

We found that combining different organic farming strategies yields significantly better results than implementing single measures. Here are some strategies for organic farming that we propose for your reference: Gradually reduce chemical dependency and transition to organic farming by combining biological control with physical weeding. For example, this combination achieves high crop yields (383.63 units) at a moderate cost (216.67dollars). Simultaneously, explore green farming and implement various organic farming scenarios.

During the transition process, extreme weather and natural disasters may impact farmland. You can collaborate with insurance companies to explore dedicated green agriculture insurance models to provide financial protection for your farmland. Additionally, you may establish green farming pilot zones in your fields to demonstrate the effectiveness of techniques such as biological control, cover crops, and organic fertilizers. These pilot zones can promote the success of green farming practices to other farmers and set an example.

Policy Advocacy Recommendations:

Organic Certification: Apply for government-supported certification (like USDA's NOP) to access high-value organic markets.

Agricultural Conservation Fund: Advocate for establishing a fund to subsidize farmers who reduce chemical use and promote biodiversity, lowering transition costs.

Technical and Financial Assistance: Take advantage of government-provided technical support and incentive programs (e.g., the U.S. Environmental Quality Incentives Program).

These programs can offer financial and technical assistance for implementing eco-friendly practices, such as optimizing farming techniques or adopting biological pest control measures.

Finally, thank you for reading our letter. We hope these suggestions are helpful to you. We believe that through scientific management, policy support, and community involvement, your farmland can achieve efficient production while becoming a model for ecological conservation.

Wishing you abundant harvests and ecological balance!

Consideration of Mature Agricultural Practices (COMAP) team

January 27th, 2025

8. Sensitivity analysis

This study systematically adjusts the parameter F, representing fertilization intensity, to examine its effects on two key ecosystem indicators—ecosystem stability and farmland status. By increasing F from 10 to 15, changes in these indicators are recorded and analyzed. Additionally, sensitivity coefficients are calculated to quantify the system's response to changes in F.

Sensitivity factor=ΔY/ΔF, ΔYdenotes the change in ecosystem indicators, ΔF is the amount of change in F.

Table 7 summarizes the results of the sensitivity analysis, presenting the calculated values of ecosystem stability and farmland status at each F value, along with their corresponding sensitivity coefficients.

Table 7. Pareto Frontier Solutions for Organic Farming Decision Making

F	10	11	12	13	14	15	Sensitivity coefficient
Ecosystem Stability	0.7541	0.7570	0.7599	0.7628	0.7656	0.7685	0.00866
Field Status	3.5156	3.4947	3.4749	3.4560	3.4380	3.4208	-0.05872

The figure reveals that the ecosystem stability indicator SS exhibits positive sensitivity to changes in FF, with an overall sensitivity coefficient of +0.00866+0.00866. This indicates that increased fertilization gradually improves ecosystem stability. The trend suggests that higher fertilization levels may enhance stability by promoting more consistent crop yields and reducing system fluctuations.

In contrast, farmland status gg shows negative sensitivity to changes in FF, with a sensitivity coefficient of −0.05872-0.05872. This implies that increased fertilization negatively affects farmland status. The declining trend in farmland status may be associated with diminishing returns of fertilization and potential adverse effects on soil quality and biodiversity.

To quantify the nonlinear impacts of various measures on ecosystem stability, we fit a set of cubic polynomial regression models to describe the dynamic relationship between input variables and ecosystem stability. The regression equation is expressed as:

$$y = b_0 + \sum_{i=1}^{n} b_i X_i + \sum_{i=1}^{n} b_{ii} X_i^2 + \sum_{i=1}^{n} \sum_{j=i}^{n} \sum_{k=j}^{n} b_{ijk} X_i X_j X_k \tag{35}$$

The fitting results are as follows:

$$y = 0.3135 + 0.0150X_1 + 0.3300X_2 + 0.0507X_1^2 + 0.8270X_2^2 + 0.0177X_1^2X_2 \tag{36}$$
$$-0.3780X_1X_2^2 - 0.3980X_1^3 + 0.4897X_2^3 - 0.0135X_1^2X_2^2 - 1.1694X_1X_2^3 + 1.0534X_1^3X_2 - 0.4302\ X_1^3X_2^2$$

The fitted cubic polynomial reflects the complex nonlinear effects of FF and its interactions on ecosystem stability hh. Simultaneously, the regression model provides predictive and explanatory capabilities, enabling the assessment of ecosystem stability changes under different combinations of measures and the optimization of management strategies.The regression results show a high coefficient of determination (R2=0.9897R^2 = 0.9897), indicating an excellent fit between the predicted and observed values. Figure 17 presents a comparison between the actual and predicted values:

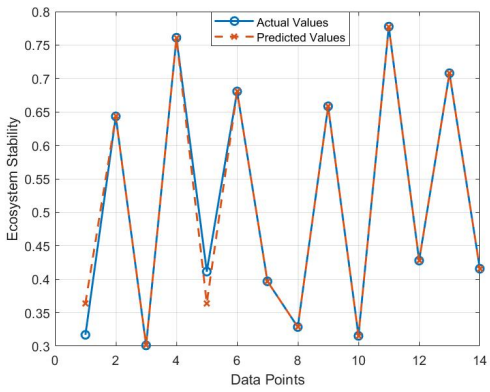


Figure 17: Comparison of actual value and predicted value

The sensitivity analysis reveals the trade-off between fertilization intensity and ecosystem stability versus farmland status. While moderate increases in F can improve ecosystem stability, excessive fertilization results in a decline in farmland status. This underscores the importance of optimizing fertilization use to balance ecosystem stability and agricultural productivity.

9. Strengths and Weaknesses

9.1 Strengths

Quantified Results and Visualization:

The model provides precise results by quantifying key indicators, making optimization analysis more scientific and actionable. Combined with visualization techniques like RGB Ecosystem CA, it intuitively demonstrates the effects of various agricultural strategies on different ecosystem components. The model effectively interprets complex multi-objective optimization results, offering decision-makers with direct and easily understandable references.

Comprehensive and Multidimensional Optimization Framework:

By integrating ecological processes and agricultural management strategies, such as chemical use and biocontrol, our model offers a holistic framework. Through Pareto optimization, the model balances multiple objectives, including crop yield, ecosystem stability, and cost. This allows stakeholders to evaluate trade-offs among different decisions and identify optimal solutions.

9.2 Weaknesses

High Data Dependency:

The model relies heavily on high-quality data, particularly for species characteristics, ecological relationship parameters, and the effects of chemical use. Insufficient or low-quality data may significantly affect the model's accuracy.

Solution: Introduce machine learning algorithms (e.g., random forest, deep learning) to estimate unknown parameters or fill in missing data.

10. Conclusion:

This study constructs a comprehensive modeling framework to explore the impacts of agricultural management strategies on ecosystems from the perspectives of spatial dynamics and ecological interactions. Based on the Lotka-Volterra predator-prey model, genetic algorithm optimization, and RGB Ecosystem CA simulations, the research highlights the critical role of moderate chemical use, biocontrol, and pollinator introduction in balancing crop yield and ecosystem stability.The results demonstrate that optimized agricultural strategies can simultaneously enhance crop yield, reduce management costs, decrease chemical dependency, and significantly improve ecosystem complexity and resilience to disturbances. By quantitatively evaluating and spatially simulating various management strategies, this study provides essential theoretical support and practical guidance for achieving the sustainable development of agricultural ecosystems.

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